



NanoPi NEO CAT1

Opta Parc Gateway · Production Deployment

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Field runbook · Bench → install → sign-off

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FriendlyWrt · Mosquitto bridge · LTE RNDIS · Opta **192.168.1.1:1883**

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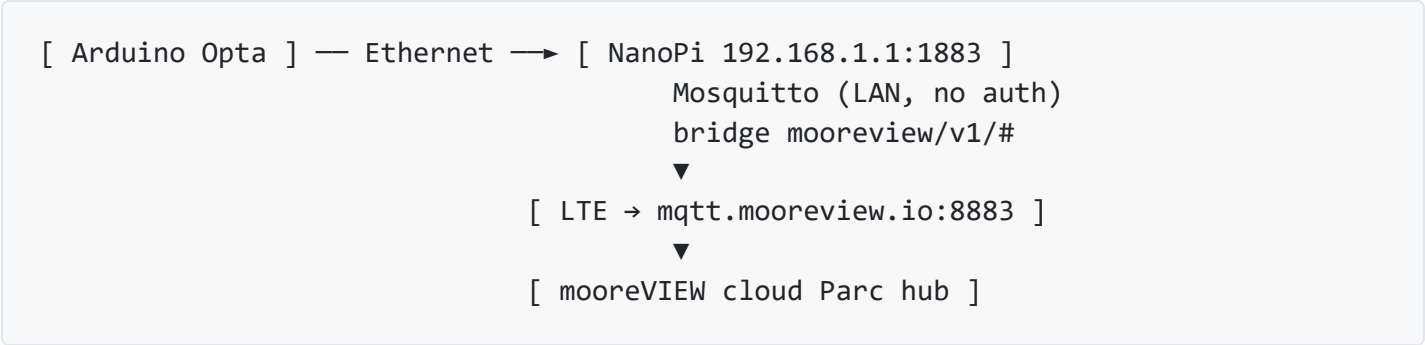
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NanoPi NEO CAT1 — production deployment runbook

Put a **mooreVIEW Opta Parc** site into production using a **NanoPi NEO + NEO CAT1** LTE module as the transparent MQTT gateway.

Doc	Use
This runbook	Bench → field → sign-off checklist
NANOPI-OPTA-GATEWAY-MQTT.md	MQTT settings (Opta + cloud)
nanopi-opta-gateway.md	Architecture and hardware notes

What you are building



- **Opta** runs ST + Parc firmware (`MooreviewOptaMqttSt`) — no change to sketch logic; only broker IP/port.
- **NanoPi** runs FriendlyWrt + Mosquitto; cloud credentials live **only on the NanoPi**.

1. Bill of materials

Qty	Item	Notes
1	NanoPi NEO (H3)	512 MB RAM; one RJ45
1	NEO CAT1 LTE module	Stacked on NEO; USB 19d1:0001
1	microSD ≥ 8 GB	Class 10
1	5 V / 3 A DC supply	Barrel or screw terminal — not weak USB
1	LTE antenna	MAIN port, tight

1	Nano SIM	Data plan activated
1	Arduino Opta	MooreviewOptaMqttSt flashed
1	Ethernet cable	Opta ↔ NanoPi eth0 (direct or via panel switch)

Optional bench: USB-TTL serial (NanoPi debug header), office LAN cable for initial SSH.

2. Software versions (validated)

Component	Version
OS image	FriendlyElec h3-sd-friendlywrt (OpenWrt 19.07, kernel 4.14)
Setup script	scripts/setup-nanopi-opta-gateway.sh
Mosquitto	1.6.15 (OpenWrt mosquitto-ssl)
Cloud broker	mqtt.mooreview.io:8883, USER mooreview

3. Bench provisioning (shop)

Complete these steps **before** shipping to site.

3.1 Flash SD card

Windows (admin PowerShell):

```
# Image: FriendlyElec h3-sd-friendlywrt-4.14-armhf-*.img.gz
# Script: scripts/flash-sd-friendlywrt.ps1 (edit $diskNumber if needed)
powershell -ExecutionPolicy Bypass -File scripts\flash-sd-friendlywrt.ps1
```

Insert SD, connect **5 V / 3 A**, antenna, SIM. For first boot you may use **eth0** → **office LAN** to find the device on DHCP.

3.2 Find the board on the office LAN

- UniFi / router client list, or
- SSH: `ssh -o HostKeyAlgorithms=+ssh-rsa -o PubkeyAcceptedAlgorithms=+ssh-rsa root@<dhcp-ip>`
Default LuCI password: try empty, then `password`.

Set a strong root password:

```
passwd
```

3.3 Install gateway software

Copy `scripts/setup-nanopi-opta-gateway.sh` to `/root/` on the NanoPi, then:

```
sed -i 's/\r$//' /root/setup-nanopi-opta-gateway.sh
chmod +x /root/setup-nanopi-opta-gateway.sh
sh /root/setup-nanopi-opta-gateway.sh install-packages
```

Get cloud MQTT password from the droplet:

```
ssh mv-mqtt
sudo grep MOSQUITTO_PASS /etc/mooreview/mqtt.env
```

On the NanoPi:

```
echo 'PASTE_MOSQUITTO_PASS_HERE' > /etc/mosquitto/cloud.pass
chmod 600 /etc/mosquitto/cloud.pass
sh /root/setup-nanopi-opta-gateway.sh configure
```

3.4 Bench tests (still on office LAN)

```
# Modem USB present
lsusb | grep 19d1

# Cellular interface (after rndis loads)
ip -4 addr show eth1
ping -c 2 -I eth1 8.8.8.8

# Mosquitto listening
netstat -ln | grep 1883

# Bridge local → cloud
sh /root/setup-nanopi-opta-gateway.sh status
```

Run the bridge test from [NANOPI-OPTA-GATEWAY-MQTT.md](#). Expect **ok** on the cloud subscriber.

Modem LEDs: STS green solid; NET **not** slow 2 s search blink (registered/idle).

3.5 Label the unit

Record on site worksheet:

- NanoPi MAC / last bench DHCP IP
- SIM ICCID
- Root password (secure store)
- Cloud tenant / site name
- Opta **deviceId** (from ATECC608 after Opta flash)

4. Opta preparation (bench)

Flash **MooreviewOptaMqttSt** (Arduino IDE). Optional compile preset — copy **opta-nanopi-gateway.defaults.example** → local defaults, add flag **-DMV_MQTT_CLOUD_PRESET=1** so first boot seeds LAN broker.

Opta **/setup** (every site — same values)

Field	Value
Ethernet	DHCP
MQTT broker	192.168.1.1
MQTT port	1883
Use TLS	off
Username / password	blank
Global site key	Match cloud project

Click **Save settings** → **Test MQTT** (only works after NanoPi is on **192.168.1.1** in the field).

5. Field installation

5.1 Wiring

[5V/3A PSU] → NanoPi power input
[LTE antenna] → CAT1 MAIN
[Opta RJ45] → NanoPi eth0 ONLY

Do not connect NanoPi eth0 to the site/building LAN router. The gateway LAN is **192.168.1.1/24** dedicated to the Opta link. Site routers at **192.168.1.1** will conflict.

5.2 Apply production network on the NanoPi

SSH to the bench IP **one last time** while eth0 is still on office LAN, **or** use serial console after moving to the field:

```
sh /root/setup-nanopi-opta-gateway.sh apply-lan
```

This sets:

Interface	Role
eth0	192.168.1.1/24 — DHCP for Opta (.50-.99)
eth1	Cellular WAN (RNDIS DHCP)

Power-cycle after apply. SSH/LuCI from a laptop on the Opta cable: **192.168.1.1** .

5.3 Power-up sequence

1. Antenna + SIM seated
2. Opta Ethernet → NanoPi eth0
3. Apply **5 V / 3 A** to NanoPi
4. Wait ~60 s — modem NET LED should leave slow search
5. Power Opta (or share supply per panel design)

6. Cloud / mooreVIEW

On the **cloud** tenant (Cloud Studio or SaaS):

1. **System setup** → **MQTT Parc hub** — enabled, broker

2. **Drivers** — add **Arduino Opta — MQTT Parc ST runtime** with the Opta `deviceId` (`opta_<serial>` from `/api/status`).
3. **Tools** → **Cellular SIMs** — link SIM ICCID to site/gateway record.
4. Deploy ST program (**Download & Start**) when ready.

7. Production acceptance tests

Check each item before leaving site.

#	Test	Pass criteria
1	NanoPi power	STS green; no reboot loop
2	Cellular	<code>ping -I eth1 8.8.8.8</code> from NanoPi (via SSH)
3	LAN broker	<code>netstat -ln grep 1883</code> on NanoPi
4	Cloud bridge	<code>netstat -tn grep 8883</code> shows ESTABLISHED
5	Opta Ethernet	Opta <code>/api/status</code> shows IP in <code>192.168.1.x</code>
6	Opta MQTT	<code>/api/status</code> → <code>mqttConnected: true</code>
7	Cloud telemetry	mooreVIEW driver Linked ; telemetry updating
8	Cloud → Opta	Download & Start or cmd reaches device
9	Global tags	Site key matches; <code>g/{key}/...</code> if used

Quick Opta check from laptop on Opta link:

```
http://<opta-dhcp-ip>/api/status
```

Quick gateway check:

```
http://192.168.1.1
ssh root@192.168.1.1
sh /root/setup-nanopi-opta-gateway.sh status
```

8. Production sign-off checklist

Print and file with the site packet.

Site name: _____ Date: _____
Technician: _____

```
[ ] SD flashed FriendlyWrt; root password set
[ ] setup-nanopi-opta-gateway.sh install-packages + configure
[ ] /etc/mosquitto/cloud.pass set (600); bridge test passed on bench
[ ] Opta flashed MooreviewOptaMqttSt; broker 192.168.1.1:1883 saved
[ ] Field wiring: Opta → eth0 only; 5V/3A supply; antenna fitted
[ ] apply-lan executed; gateway reachable at 192.168.1.1
[ ] Modem registered (NET not slow-searching > 3 min)
[ ] Opta mqttConnected; cloud telemetry live
[ ] SIM linked in mooreVIEW Cellular SIMs
[ ] ST program deployed and running (if applicable)
```

Notes: _____

9. Troubleshooting

Symptom	Likely cause	Fix
NET slow ~2 s blink	Weak power, bad SIM, no signal	3 A PSU ; reseal SIM/antenna; check data plan
No eth1	RNDIS driver not loaded	<code>modprobe rndis_host</code> ; ensure <code>/etc/modules.d/rndis</code> exists; reboot
Bridge auth failed	Wrong <code>cloud.pass</code>	Re-copy <code>MOSQUITTO_PASS</code> ; <code>sh .../configure</code>
Opta no MQTT	Wrong broker or not on Opta LAN	Broker 192.168.1.1:1883 ; Opta must DHCP from NanoPi
Can't SSH after deploy	Still on office LAN / IP conflict	Use 192.168.1.1 on direct cable; unplug from site router
SSH "no matching host key"	Old OpenWrt RSA key	<code>ssh -o HostKeyAlgorithms=+ssh-rsa ...</code>
Telemetry in cloud, no cmd down	Bridge topic or hub config	Confirm <code>mooreview/v1/# bridge</code> ; cloud Parc hub enabled

Logs on NanoPi:

```
logread | grep -iE 'mosquitto|bridge|network'
/etc/init.d/mosquitto restart
/etc/init.d/network restart
```

10. Security (production)

- Change **root** password from factory default (`passwd`).
- Store **MOSQUITTO_PASS** only on the NanoPi (`/etc/mosquitto/cloud.pass`) — never on Opta.
- Do not commit passwords to git; use [nanopi-gateway-cloud.env.example](#) locally only.
- Restrict physical access to the panel; NanoPi LAN has no MQTT auth by design (Opta-only link).

11. Related files

Path	Purpose
<code>scripts/setup-nanopi-opta-gateway.sh</code>	Provision script
<code>scripts/flash-sd-friendlywrt.ps1</code>	Windows SD flash
<code>deploy/cloud/phase1/NANOPI-OPTA-GATEWAY-MQTT.md</code>	MQTT reference
<code>deploy/cloud/phase1/opta-nanopi-gateway.defaults.example</code>	Opta compile preset
<code>deploy/cloud/phase1/OPTA-CLOUD-MQTT.md</code>	Direct cloud Opta (not this gateway)
<code>docs/training/reference/hardware/cellular-opta-gateway.md</code>	LilyGO equivalent

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Revision: 2026-08-11 — field stack NanoPi NEO + NEO CAT1 + FriendlyWrt + Mosquitto bridge.